Lab 4

Create NetBeans JavaFX Project

- File → New Project → JavaFX → JavaFX Application
- Name: `RegistrationFormApp`

JavaFX Structure

```
public class Main extends Application {
    @Override
    public void start(Stage primaryStage) {
        // UI code here
    }
    public static void main(String[] args) {
        launch(args);
    }
}
```

Common Controls

- `Label` – text display
- `TextField` – single line input
- `PasswordField` – password input
- `RadioButton` + `ToggleGroup` – single selection
- `CheckBox` – multiple selection
- `ComboBox` – dropdown
- `DatePicker` – date selection
- `Button` – click action
- `TableView` – display records

Event Handling

```
button.setOnAction(e -> {
    // action code
});
```

5. Alert Boxes

```
Alert alert = new Alert(Alert.AlertType.INFORMATION);
alert.setContentText("Message");
```

```
alert.show();
```

LESSON 1: Hello World JavaFX

Goal: Understand basic JavaFX structure and create a window.

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Label;
import javafx.scene.layout.StackPane;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label message = new Label("Welcome to Registration Form");
        StackPane root = new StackPane(message);
        Scene scene = new Scene(root, 400, 300);

        primaryStage.setTitle("JavaFX App Title");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Task: Change window size to 600x400 and title to `"Registration Form"`

LESSON 2: Adding TextField and Button

Goal: Learn basic input controls and button click events.

Code:

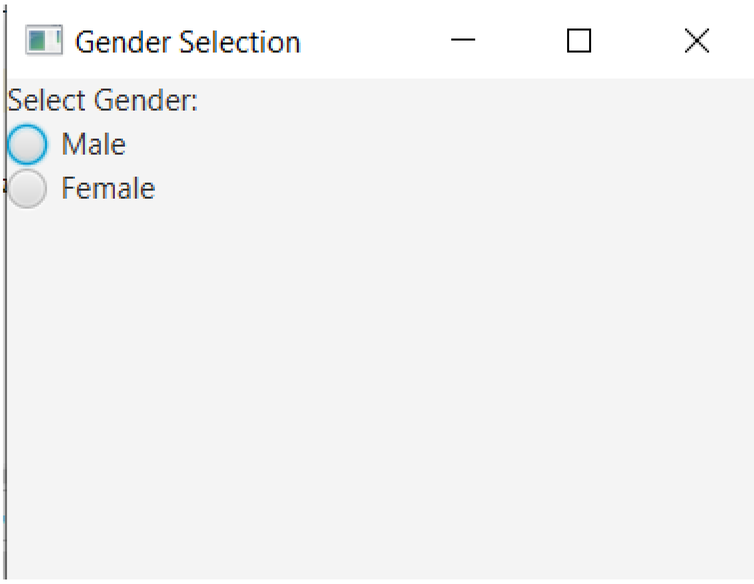
```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.TextField;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;
```

```java
public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label nameLabel = new Label("Name:");
        TextField nameField = new TextField();
        Button submitButton = new Button("Submit");
        Label resultLabel = new Label();

        submitButton.setOnAction(e -> {
            String name = nameField.getText();
            resultLabel.setText("Hello " + name);
        });

        VBox root = new VBox(10, nameLabel, nameField, submitButton,
resultLabel);
        Scene scene = new Scene(root, 400, 300);

        primaryStage.setTitle("Registration Form");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Task: Add a second TextField for "Email" and display both name and email when button is clicked.

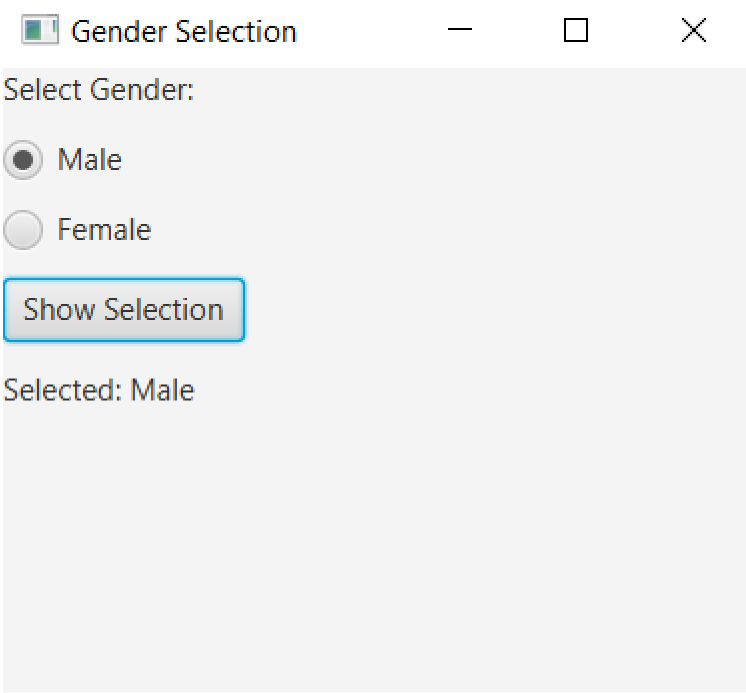
LESSON 3: Working with RadioButtons

Goal: Learn how to use RadioButton and ToggleGroup for single selection.



Code 3.1:

```
public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label genderLabel = new Label("Select Gender:");
        RadioButton maleRadio = new RadioButton("Male");
        RadioButton femaleRadio = new RadioButton("Female");

        VBox root = new VBox(genderLabel,maleRadio,femaleRadio);

        Scene scene = new Scene(root,300,200);
        primaryStage.setTitle("Gender Selection");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```



Code 3.2:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.RadioButton;
import javafx.scene.control.ToggleGroup;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label genderLabel = new Label("Select Gender:");

        RadioButton maleRadio = new RadioButton("Male");
        RadioButton femaleRadio = new RadioButton("Female");

        ToggleGroup genderGroup = new ToggleGroup();
        maleRadio.setToggleGroup(genderGroup);
        femaleRadio.setToggleGroup(genderGroup);

        Button showButton = new Button("Show Selection");
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            if (maleRadio.isSelected()) {
                resultLabel.setText("Selected: Male");
            } else if (femaleRadio.isSelected()) {
```

```
                resultLabel.setText("Selected: Female");
            } else {
                resultLabel.setText("Nothing selected");
            }
        });

        VBox root = new VBox(10, genderLabel, maleRadio, femaleRadio, showButton,
resultLabel);
        Scene scene = new Scene(root, 300, 250);

        primaryStage.setTitle("Gender Selection");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Learn more: `ToggleGroup` , `isSelected()` method

LESSON 4: Working with CheckBoxes

Goal: Learn CheckBox for multiple selection.

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.CheckBox;
import javafx.scene.control.Label;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label techLabel = new Label("Technologies Known:");

        CheckBox javaCheck = new CheckBox("Java");
        CheckBox dotnetCheck = new CheckBox("DotNet");
        CheckBox pythonCheck = new CheckBox("Python");

        Button showButton = new Button("Show Selected");
```

```
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            String selected = "";
            if (javaCheck.isSelected()) selected += "Java ";
            if (dotnetCheck.isSelected()) selected += "DotNet ";
            if (pythonCheck.isSelected()) selected += "Python ";

            if (selected.isEmpty()) {
                resultLabel.setText("No technology selected");
            } else {
                resultLabel.setText("Selected: " + selected);
            }
        });

        VBox root = new VBox(10, techLabel, javaCheck, dotnetCheck, pythonCheck,
showButton, resultLabel);
        Scene scene = new Scene(root, 300, 250);

        primaryStage.setTitle("Technologies Selection");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Task: Add a "Select All" button that selects all checkboxes.
Learning: CheckBox, multiple selections, string concatenation

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LESSON 5: Working with ComboBox

Goal: Learn ComboBox for dropdown selection.

Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.Label;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {
```

```java
    @Override
    public void start(Stage primaryStage) {
        Label qualLabel = new Label("Educational Qualification:");

        ComboBox<String> qualCombo = new ComboBox<>();
        qualCombo.getItems().addAll("Engineering", "MCA", "MBA", "BCA");
        qualCombo.setPromptText("Select qualification");

        Button showButton = new Button("Show Selection");
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            String selected = qualCombo.getValue();
            if (selected == null) {
                resultLabel.setText("Nothing selected");
            } else {
                resultLabel.setText("Selected: " + selected);
            }
        });

        VBox root = new VBox(10, qualLabel, qualCombo, showButton, resultLabel);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Qualification Selection");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Student Task: Add a default value "Engineering" that appears automatically when the app starts.

What they learn: ComboBox, getItems(), addAll(), setValue()

LESSON 6: Using DatePicker

Goal: Learn DatePicker for date input.

Code:

```java
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
```

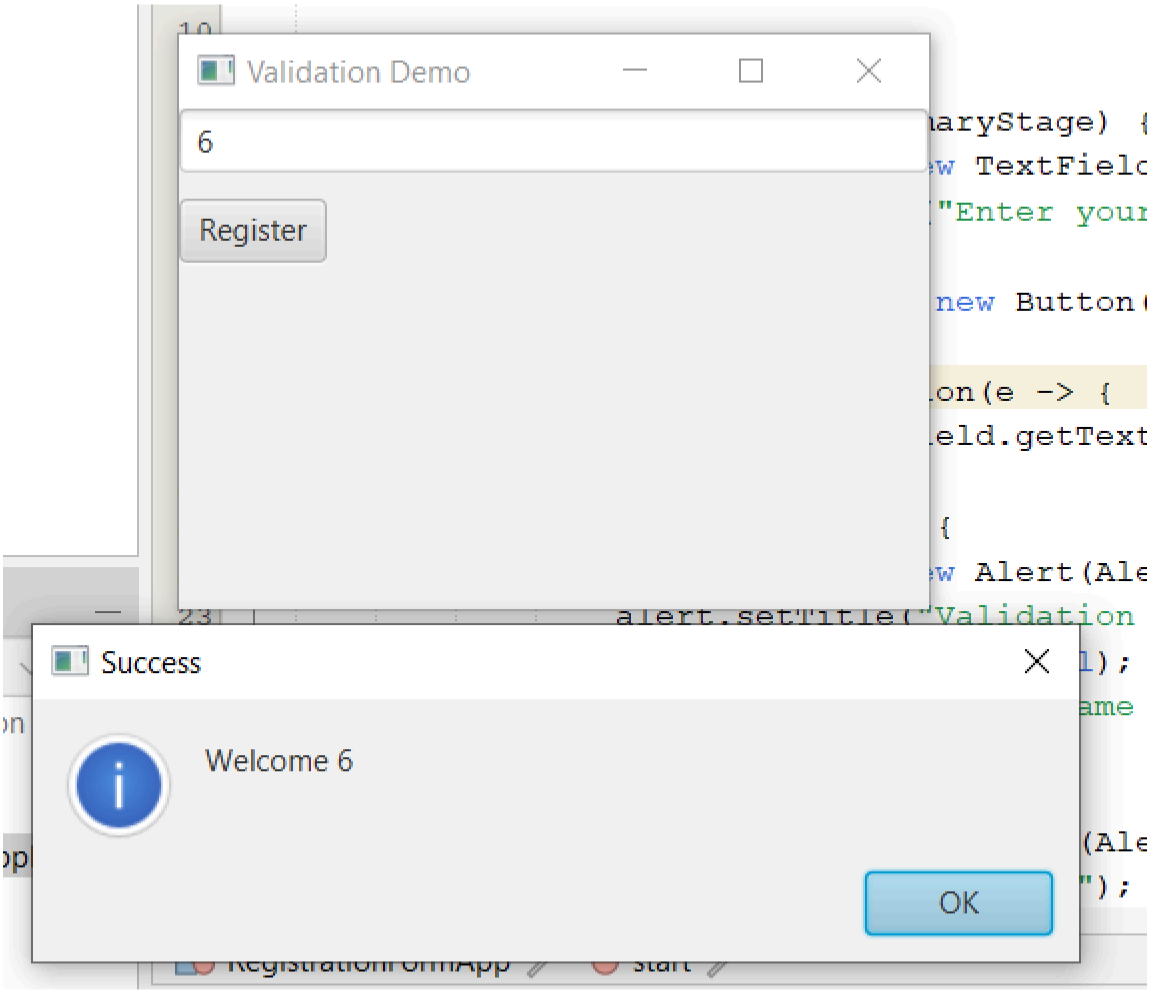
```java
import javafx.scene.control.DatePicker;
import javafx.scene.control.Label;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;
import java.time.format.DateTimeFormatter;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        Label dobLabel = new Label("Date of Birth:");
        DatePicker dobPicker = new DatePicker();

        Button showButton = new Button("Show Date");
        Label resultLabel = new Label();

        showButton.setOnAction(e -> {
            if (dobPicker.getValue() == null) {
                resultLabel.setText("No date selected");
            } else {
                String formattedDate =
dobPicker.getValue().format(DateTimeFormatter.ofPattern("dd/MM/yyyy"));
                resultLabel.setText("DOB: " + formattedDate);
            }
        });

        VBox root = new VBox(10, dobLabel, dobPicker, showButton, resultLabel);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Date Selection");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Student Task: Display the date in "yyyy-MM-dd" format instead.

Learing: DatePicker, DateTimeFormatter, getValue()

LESSON 7: Alert Boxes and Validation

Goal: Learn to show popup messages and validate input.



Code:

```
import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.TextField;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;

public class RegistrationFormApp extends Application {

    @Override
    public void start(Stage primaryStage) {
        TextField nameField = new TextField();
        nameField.setPromptText("Enter your name");

        Button registerButton = new Button("Register");

        registerButton.setOnAction(e -> {
            String name = nameField.getText().trim();
            // Learn what .trim() does
            if (name.isEmpty()) {
                Alert alert = new Alert(Alert.AlertType.ERROR); // Learn more
```

```java
                alert.setTitle("Validation Error");             // about AlerType
                alert.setHeaderText(null);
                alert.setContentText("Name cannot be empty!");
                alert.showAndWait();
            } else {
                Alert alert = new Alert(Alert.AlertType.INFORMATION);
                alert.setTitle("Success");
                alert.setHeaderText(null);
                alert.setContentText("Welcome " + name);
                alert.showAndWait();
            }
        });

        VBox root = new VBox(10, nameField, registerButton);
        Scene scene = new Scene(root, 300, 200);

        primaryStage.setTitle("Validation Demo");
        primaryStage.setScene(scene);
        primaryStage.show();
    }

    public static void main(String[] args) {
        launch(args);
    }
}
```

Task: Add age validation (age must be between 18 and 60).

Learning: `Alert class` , `AlertType.ERROR` , `AlertType.INFORMATION` , input validation
